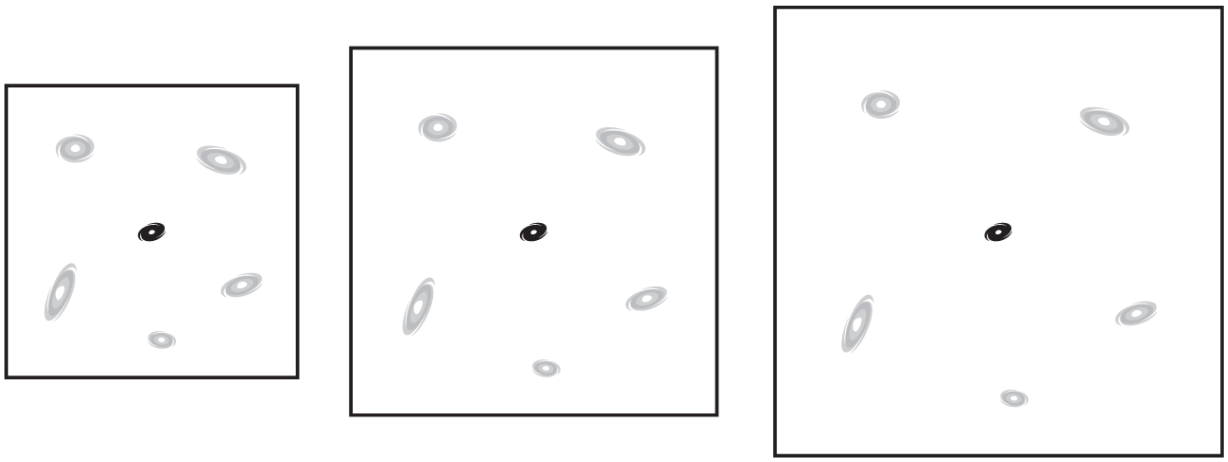


Figure 7: Illustration of cosmic expansion. Here, the increasing size of the universe at three different moments is represented by the increasing size of the square from left to right. As time goes on, galaxies move apart from one another, but they do not get bigger with the expansion of space.

The upshot is that every point in the universe is the nexus of what appears to be a powerful uniform repulsion. Technically, the universe doesn't have a center. But we're each the center of our own *observable* universe.[fn1] And from our perspective, all the galaxies farther out than our near neighbors are careening away from us as fast as they can. It's not us; it's cosmology.

Cosmic expansion was harder to discover than you might think. While galaxies other than our own have been visible through telescopes since the 1700s, their distances are so great and motions so slow (as judged by human timescales) that determining how they were moving relative to us, and even that they were galaxies at all, took more than two centuries. Even now, the most powerful telescopes can't see the motion directly—the galaxies don't appear to be farther away every time we look. But we can detect it by carefully teasing it out of a seemingly unrelated property of galaxies: the color of their light.

If you've ever heard the rising and suddenly falling "vroo-oom!" sound of a race car passing by, or the shift in tone when a siren approaches and retreats, you're familiar with the Doppler effect. A Doppler shift of the sort you normally experience is the phenomenon of a sound getting higher pitched as the object emitting it moves toward you, and dropping in pitch when it moves away. It has to do with the way the pressure waves in the air pile up on approach and stretch out on departure, changing the frequency of the sound that you hear. Frequency is, after all, just how quickly the waves